Supplemental Material. Articles relating to gluteal tendinopathy, treatment and main findings

Lead Author (Year)	LoE	Intervention	Pathology tendinopathy (distribution)	N	Sex, F/M	Age, y, mean (range)	Preop DoS, mo mean (range)	Follow-up, mo, mean (range)	Reported outcome measure	Outcome assessment	Initial / follow up mean outcome scores (range)	Complications / SAEs	Key to Symbols
Blakey (2020)[4]	1b	Endoscopic bursectomy and ITB release with/without radiofrequency microdebridement	1-3	33	30/3	58 (SD ±8.3)	NR (minimum 6)	12	mHHS (no RF) mHHS (RF) PASS (no RF) at 1 y PASS (RF) at 1 y △WOMAC (no RF) △WOMAC (RF) △VAS (no RF) △VAS (RF) △SF-12 (no RF) △SF-12 (RF)	BL / 6mo / 12mo	57.5 (±10.6) / 77.4 (±20.0)** / 77.8 (±18.4)** 59.0 (±12.3) / 71.8 (±17.8)*** / 80.0 (±18.9)** 63%* 69%* 13.2**° 27.0**° 3.0**° 3.6**° 8.3****° 11.3****°	2 patients underwent surgery to the hip joint within follow up; no surgery-related SAEs	*P<0.01 to BL **P<0.01 to BL ***P<0.05 to BL ****P=ns °RF vs no RF P=ns
Fitzpatrick (2019)[26]; Fitzpatrick (2018)[25]	1b	1. US-guided PRP injection vs 2. US-guided CS injection	1, 2, 3ᶜ PRP: (50%/15%/35) CS: (60%/22.5%/17.5%)	80	72/8	60 (23-78)	15 (SD ±12.35)	NR (minimum 24)	mHHS (PRP) mHHS (CS) Recovery (PASS >74) (PRP) Recovery (PASS >74) (CS)	BL / 6w / 3mo / 6mo / 12mo / 24mo BL / 3mo / 6mo / 12mo / 24mo	53.7(±12.1) / 68.8(±13.3) / 74.1(±13.9)* / 77.6(±11.9)** / 78.2(±14.5)** / 82.6(±9.7)** 54.2(±10.9) / 69.5(±14.8) / 67.1(±16.0)* / 65.7(±15.3)** / 70.5(±23.8)** / 71.3(±25.8)** na / 64.1% / 65.8% / 76.3% / 88.6% na / 45.9% / 27.0% / 60.0% / 80.0%	no SAEs	*PRP vs CS P<0.05 **PRP vs CS P<0.001 ᶜ tendinopathy grade 4 excluded
Carlisi (2019)[10]	1b	1. ESWT vs 2. US therapy	2-3ᶜ	50	44/6	61.24 (SD ±9.26)	6 (3-12)	NR	NRS (SWT) NRS (US) LEFS (SWT) LEFS (US)	BL / 2mo / 6mo	5.1 (±2.2) / 2.1 (±2.1)°* / 0.8 (±1.3)°* 4.9 (±1.7) / 3.4 (±2.1)°* / 2.0 (±2.0)°* 49 (±19.1) / 65.1 (±11.2)° / 68.2 (±11.5)° 47.2 (±17.9) / 57.5 (±11.9)° / 63.4 (±13.0)°	no SAEs	°P<0.001 to BL *SWT vs US P<0.05 ᶜ tendinopathy grades 1 and 4 excluded
Mellor (2018)[42]	1b	1. US-guided CS injection vs 2. PHT with registered therapist vs 3. no specific treatment	2	204	167/37	54.8 (SD ±8.8)	24 (8-48)	NR (minimum 12)	GRCS (success) (EDX) GRCS (CS) GRCS (WS) NRS (EDX) NRS (CS) NRS (WS)	BL / 1mo / 2mo / 3mo / 6mo / 12mo	na / 58%°° / 77%°°* / 74%°° / 74%°°* / 78%°°* na / 58%ºº / 58%*ºº / 60%º / 53%* / 57%* na / 20%ºº°° / 29%ºº°° / 33%°°° / 38%°° / 52%°° 4.9(±1.0) / 2.7(±1.6)°° / 1.5(±1.5)°°* / 1.7(±1.7)°° / 1.8(±1.9)° / 2.1(±2.2)° 4.8(±1.0) / 2.4(±2.0)ºº / 2.7(±2.4)ºº* / 2.5(±2.1)º / 2.6(±2.2) / 2.3(±1.9) 4.8(±1.0) / 4.1(±1.9)°°ºº / 3.8(±2.0)°°ºº / 3.6(±2.2)°°º / 3.1(±2.4)° / 3.2(±2.6)°	NR	°EDX vs WS P<0.05 °°EDX vs WS P<0.001 *EDX vs CS P<0.05 ºCS vs WS P<0.05 ººCS vs WS P<0.001
Jacobson (2016)[33]	1b	1. US-guided PRP injection vs 2. US-guided tendon fenestration	2ᶜ	30	24/6	57 (24-81)	NR	3	VAS (PRP) VAS (TF)	BL / 1w / 2w	31.4 (±7.3) / 25.5 (±8.8)** / 19.4 (±10.3)** 32.4 (±10.2) / 16.8 (±11.5)** / 15.2 (±10.8)**	NR	**P<0.001 to BL PRP vs FT P = ns ᶜ tendinopathy grade 4 excluded
Saltzman (2018)[50]	3	Open or endoscopic tendon repair (SA: SR [small or medium tears], DR [large or massive tears]) 1. with PRFM augmentation vs 2. without PRFM augmentation	3, 4	47	41/6	62.0 (37.8-88.5)	31 (2-240)	31.3 (SD ±12)	VAS (PRFM) VAS (Co) HOS ADL (PRFM) HOS ADL (Co) HOS Sport (PRFM) HOS Sport (Co mHHS (PRFM) mHHS (Co) HHS (PRFM) HHS (Co) SF-12 Physical (PRFM) SF-12 Physical (Co) Patient satisfaction (PRFM) Patient satisfaction (Co) iHOT-12 (PRFM) iHOT-12 (Co)	BL / mean FU	7.2(±2.0) / 2.5(±2.6)** 6.4(±1.7) / 1.9(±2.1)** 53.7(±13.9) / 82.7(±15.4)** 57.8(±11.2) / 82.79(±11.8)** 30.4(±21.9) / 68.0(±26.4)** 37.2(±12.7) / 72.3(±18.0)** 50.9(±11.1) / 77.0(±19.9)** 56.0(±6.1) / 78.7(±13.5)** NR / 73.6(±18.9) NR / 77.2(±13.3) NR / 45.5(±10.1)° NR / 28.1(±8.3)° NR / 82.1(±31.6) NR / 89.3(±13.5) NR / 64.6(±31.9)° NR / 44.0(±24.0)°	4 re-tears	**P<0.001 to BL °PRFm vs Co P<0.05
Ebert (2019)[22]	4	Open augmented tendon repair (TS, SA, LARS), V-Y lengthening of ITB	3, 4	84	84/0*	64.6 (43-84)	40.8 (6-240)	24	HHS OHS VAS frequency VAS severity SF-12 PCS SF-12 MCS 6MWT (m) Strength (kg) Patient satisfaction	BL / 24mo	58.6 (±17.0) / 92.5 (±9.6)** 25.6 (±7.5) / 43.3 (±5.4)** 6.8 (±2.1) / 1.4 (±1.6)** 8.4 (±2.2) / 1.5 (±2.0)** 33.7 (±7.9) / 45.3 (±9.6)** 48.3 (±11.3) / 55.2 (±9.4)** 416 (±91) / 508 (±84)** 9.3 (±4.5) / 10.3 (±4.7)** 96.4% satisfied	No RtT, 3 retears, 2 superficial wound infection, 1 hematoma	**P<0.001 to BL *male participants excluded

Lead Author (Year)	LoE	Intervention	Pathology tendinopathy (distribution)	N	Sex, F/M	Age, y, mean (range)	Preop DoS, mo mean (range)	Follow-up, mo, mean (range)	Reported outcome measure	Outcome assessment	Initial / follow up mean outcome scores (range)	Complications / SAEs	Key to Symbols
Ebert (2018)[21]	4	Open augmented tendon repair (TS, SA, LARS), V-Y lengthening of ITB	3, 4	110	101/9	63.2 (43-82)	42 (6-216)	NR (minimum 12)	HHS OHS SF-12 PCS SF-12 MCS VAS frequency VAS severity 6-min walk distance (m) Isometric ABD strength (kg) GRCS Satisfaction	BL / 3m / 6mo / 12mo 12 mo	57.6(±18.9) / 74.5(±16.1) / 80.8(±14.9) / 85.8(±13.6)** 25.3(±8.7) / 33.8(±8.4) / 37.3(±7.9) / 39.9(±6.7)** 33.2(±9.2) / 36.4(±10.5) / 40.6(±10.1) / 44.1(±9.4)** 49.3(±11.5) / 53.5(±11.4) / 52.4(±11.4) / 54.9(±9.8)* 7.9(±2.6) / 3.8(±2.8) / 3.1(±2.6) / 2.2(±2.0)** 6.5(±2.2) / 2.8(±1.8) / 2.7(±2.1) / 1.9(±1.5)** 405(±109) / 417(±112) / 460(±103) / 495(±102)** 35.7(±7.2) / 36.2(±8.2) / 38.8(±8.0) / 39.9(±8.6)* 3.5 (1-5) 96% satisfied	3 surgical failures (2RtT), 1 hematoma, 2 superficial wound infection (AB), 1DVT/PE	*P<0.01 to BL **P<0.0001 to BL
Hartigan (2018)[29]	4	Endoscopic tendon repair (SA)	3, 4	25	24/1	53.5 (38.4-70.7)	NR	38 (26.6-68)	mHHS HOS ADL HOS Sport NAHS VAS	BL / mean FU	54.9 / 76.2** 50.2 / 80.6** 30.9 / 67.3** 51.9 / 82.4** 7.1 / 2.7**	no complications	**P<0.001 to BL
McGonagle (2018)[41]	4	Open tendon repair (SA), V-Y lengthening of ITB if necessary	3, 4	14 (15 hips)	10/4	63 (49-82)	NR	NR	HHS Trendelenburg sign Patient satisfaction	BL / FU (very sat/sat/neutral /dissatisfied)	NR / 66(35-88) 86.7% / 50% 26.7% / 13.3% / 26.7% / 33.3%	NR	*P* NR
Seo (2018)[51]	4	ESWT	2 (83.3%) - 3 (16.7%)	18	10/8	59 (31-78)	33 (6-180)	27	NRS Success (RMS "good" or "excellent")	BL / 1w / 27mo	5.9(±1.6) / 2.5(±1.5)* / 3.3(±3.0)* na / 83.3% / 55.6%	no AE	*P<0.01 to BL
Thaunat (2018)[53]	4	Endoscopic tendon repair (SA: SR [partial-thickness tears], DR [full-thickness tears])	3 (68.2%), 4 (31.8%)	20 (22 hips)	17/3	66 (SD ±8)	42 (SD ±38.4)	31.7 (24-47)	VAS mHHS NAHS Patient satisfaction (1-10)	BL / mean FU	7.2(±1.0) / 3.2(±2.8)* 33.7(±8.7) / 80.2(±19.4)** 47.7(±7.7) / 76.8(±14.5)** na / 7.9(±2.3)	1 failure of repair (no RtT)	*P<0.05 to BL **P<0.001 to BL
Bucher (2017)[9]	4	Autologous tenocyte injection	1, 2, 3	12	12/0	52.6 (41-67)	33.2 (6-144)	8.7	OHS VAS SF-36 PCS Satisfaction	BL / 3mo / 6mo / 12mo / 24mo 12mo	24.0 / NR / 32.3* / 38.9** / NR 7.4 / 4.6** / NR / 3.4** / 2.9** 28.1 / NR / NR / 43.3** / 40.9** 5/12 highly satisfied, 3/12 satisfied	no SAEs, 3 donor site discomfort	*P<0.05 to BL **P<0.001 to BL
Huxtable (2017)[31]	4	Open augmented tendon repair (TS, SA, LARS), V-Y lengthening of ITB	3, 4	21	19/2	62.1 (44-78)	49.2 (12-120)	NR (minimum 12)	HHS OHS SF-12 PCS SF-12 MCS VAS frequency VAS severity 6-min walk distance (m) ABD strength (%BW)	BL / 6mo / 12mo	65.2(±19.3) / 83.6(±12.4) / 92.2(±7.3)** 28.8(±8.2) / 40.5(±5.5) / 41.2(±4.1)** 36.7(±9.5) / 44.0(±8.6) / 45.5(±7.9)** 54.6(±10.6) / 57.0(±9.1) / 59.2(±5.4)** 7.3(±2.8) / 2.2(±2.2) / 2.0(±2.0)** 6.2(±2.6) / 2.2(±1.8) / 1.8(±1.4)** 446.3(±79.2) / 47.6(±67.3) / 532.9(±67.5)** 13.6(±3.8) / 17.3(±6.9) / 17.4(±6.4)**	NR	**P<0.0001 to BL
Coulomb (2016)[14]	4	Endoscopic tendon debridement and microperforation, no tendon repair	2, 3	17	16/1	53.5 (17-71)	34.8 (6-108)	37.6 (20-62)	VAS UCLA activity score HHS Patient satisfaction (very sat/ sat/disappointed/dissatisfied)	BL / mean FU	7.2(±1.1) / 3.3(±1.9)** na / 45% resumed sports participation na / 6.2(±2.4) 0 / 11 / 4 / 2	no major complications	**P<0.001 to BL
Lee (2016)[37]	4	US-guided PRP injection and concomitant gluteus medius needle tenotomy	1, 2, 3	21	17/4	48 (25-68)	NR (minimum 3)	19.7 (12.1-32.3)	mHHS HOS ADL HOS Sport iHOT-33	BL / mean FU	56.7(±11.2) / 74.2(±15.07)** 68.9(±16.5) / 84.1(±12.4)** 45.5(±23.4) / 66.7(±24.6)* 34.1(±15.3) / 66.3(±23.1)**	no SAEs	*P<0.001 to BL **P<0.0001 to BL
Jacobson (2015)[32]	4	US-guided fenestration of gluteus medius or minimus tendon	2 (84.6%), 3 (15,4%)[ς]	13	NR	55.8 (26.6-77)	NR	2.3 (0.3-27)	Patient satisfaction (improvement)	marked / some / no change / deterioration	53.8% / 23.1% / 15.4% / 7.7%	NR	*P* NR [ς] tendinopathy grade 4 excluded
Bucher (2014)[8]	4	Open augmented tendon repair (TS, SA, LARS), V-Y lengthening of ITB	3, 4	22	19/3	62 (49-74)	35 (4-96)	NR (minimum 12)	OHS VAS SF-36 PCS SF-36 MCS Patient satisfaction (high sat / sat / quite sat)	BL / 3m / 6mo / 12mo	22.4(±7.3) / 35.0(±7.7)** / 38.3(±6.3)** / 41.1(±6.5)** 7.1(±1.3) / NR / NR / 1.6(±1.4)** 29.7(±6.4) / NR / NR / 44.4(±11.3)** 45.8(±11.5) / NR / NR / 52.5(±8.6)* 54.5% / 18.2% / 27.3%	1 RtT for irritating screw head	*P<0.01 to BL **P<0.001 to BL

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Makridis (2014)[38]	4	Open tendon repair (SA: DR)	3, 4	67	62/5	68 (25-87)	33.6 (6-120)	55.2 (12-96)	VAS VAS (atrophic) VAS (nonatrophic) Lequesne Index Lequesne index (at) Lequesne index (non-at) HHS HHS (atrophic) HHS (nonatrophic) Significant disability (%) Sig. disability (AT)(%) Sig. disability (non-AT)(%) Pain on stairs (%) Pain on stairs AT)(%) Pain on stairs (non-AT)(%)	BL / mean FU	8.7(±1.1) / 1.7(±2.7)** 8.2(±1.1) / 3.1(±2.9)° 8.9(±1.1) / 1.3(±1.8)° 12.3(±2.6) / 4.0(±4.9)** 12.9(±2.8) / 7.4(±3.9)°° 12.3(±2.6) / 3.2(±3.5)°° 50.4(±8.0) / 87.9(±15.5)** 51.2(±10.5) / 75.7(±19.7)°° 49.3(±7.5) / 90.7(±12.7)°° 94% / 6%** 90.5% / 21%° 95% / 2.5%° 83% / 11.5%** 78.6% / 21% 89.5% / 8%°	11 retears (16%) (all RtT)	**P<0.001 to BL °AT vs non-AT P<0.05 °°AT vs non-AT P=0.001
Davies (2013)[18]	4	Open tendon repair (SA [small tears], TS ± allograft [large tears])	3, 4	22	20/2	67.7 (45-85)	NR	70.8 (61-100)	HHS LEAS Muscle strength (0-5) "Undergo surgery again?" improvement in function improvement in pain	BL / 12mo / 60mo	53(±10.9) / 87(±16.4)** / 88(±11.5)** 6.7(±0.5) / 8.9(±1.3)** / 8.8(±0.9)** 3.1(±1.0) / 4.7(±0.7) / NR 84.2% 79.6% (20%-100%) 90% (40%-100%)	2 retears following falls, no other complications	**P<0.00001 to BL
Mautner (2013)[39]	4	US-guided PRP injection (single or multiple)	2-3	16	NR	48 (19-73)	36 (6-360)	15 (SD ±6)	△VAS Perceived improvement	mean FU	`-5.2* (81% moderate improvement to complete resolution of symptoms)	NR	*P<0.001 to BL
McCormick (2013)[40]	4	Endoscopic tendon repair (SA)	4	11	7/3	65.9 (60-74)	11.7 (1.5-40.9)	22.6 (12.6-38.7)	mHHS HOS ADL HOS Sport “Undergo surgery again?" ABD strength (5-pt scale)	BL / mean FU	NR / 84.7(±14.5) NR / 89.1(±11.3) NR / 76.8(±31.0) na / 90% 3.3 / 4.6(±0.66)	NR	*P* NR
Walsh (2011)[55]	4	Open tendon repair (TS)	3, 4	72	67/5	62 (36-88)	22.4 (6-144)	NR (minimum 12)	Merle D'Aubigne Postel score Gait Distance	BL / 6mo / 12 mo	10.9(±0.3) / 16.6(±0.3**) / 16.7(±0.4)** 0.8(±0.1) / NR / 5.2(±0.2)** 4.7(±0.2) / NR / 5.6(±0.1)**	6 DVT, 4 retear, 3 hematoma, 1 PE, 1 fractured GT, 1 infection, 1 pressure sore	**P<0.0001 to BL
Fearon (2010)[23]	4	Open tendon repair (TS)	3, 4	24	24/0	56 (38-76)	33.8 (3-120)	22 (12-56)	VAS HHS ODI Patient satisfaction	BL / mean FU	8.5 (7.5-10) / 0.9 (0-2.5)** NR / 74 (59-85) NR / 82 (74-92) na / 78.6/100 (50-100)	NR	**P<0.001 to BL
Labrosse (2010)[36]	4	US-guided single CS injection	1 (11%), 2 (87%), 3 (12%)	54	48/6	54.7 (35-78)	NR	1	VAS Patient satisfaction	BL / 1mo	6.4 / 2.9** 70% satisfied with result	NR	**P<0.001 to BL
Davies (2009)[17]	4	Open tendon repair (SA, DR)	3, 4	16	15/1	63 (47-82)	23 (6-48)	12	VAS Merle D'Aubigne Postel score OHS SF-36 PCS SF-36 MCS	BL / mean FU	7.0 / 2.0* 10.5 / 15.0 ** 21.4 / 38.9** 28.4 / 40.2 ** 54.9 / 59.4	4 retears (3RtT), 1 deep infection (RtT)	*P<0.01 to BL **P<0.001 to BL

LoE Level of evidence, N sample size, DoS duration of symptoms, BL baseline, NR not reported, FU follow-up, AE adverse events, SAE severe adverse events, F female, M male, mo months, ns not significant, Sig significant, pt point, ABD abduction, RtT return to theatre, TS transosseous sutures, SA suture anchors, DR double row, GT greater trochanter, DVT deep vein thrombosis, PE pulmonary embolism, AT atrophic, non-AT non atrophic, US ultrasound, CS corticosteroids, PRP platelet rich plasma, VAS Visual Analog Scale, SF-36 Short Form 36, PCS Physical Component Score, MCS Mental Component Score, (m)HHS (modified) Harris Hip Score, OHS Oxford Hip Score, LEAS Lower Extremity Activity Sale, ODI Oswestry Disability Index, HOS Hip Outcome Score, ADL activities of daily living.